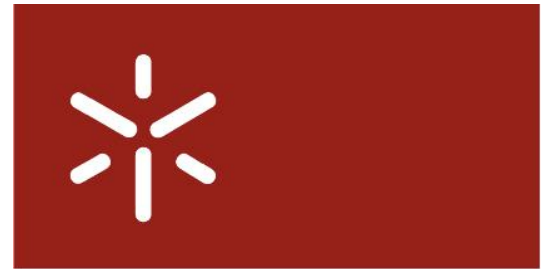

Mobile Networks

2024/2025

Project description



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Context (i)

Radio frequency spectrum is one of the most scarce resources in telecommunications.

Efficient use of the spectrum requires the use of advanced modulation and coding methods to:

- provide high data rate services
- offer services to an increasing number of users
- increase network capacity

Context (ii)

One of the methods used to provide multiple access in radio-based telecommunication networks is CDMA – Code Division Multiple Access.

CDMA allows multiple users to share the same frequency band simultaneously.

The challenge

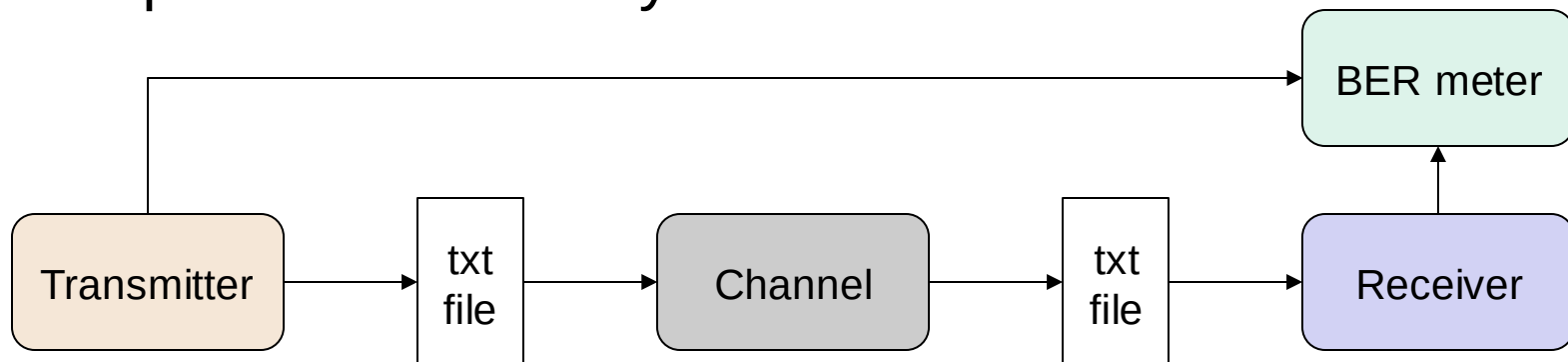
The challenges of this project are:

- to transmit data from multiple users sharing the same radio channel
- maximize the number of simultaneous users that can share the channel with acceptable performance

Approach

The proposed approach is based on creating a simulator of CDMA, including:

- a transmitter
- a noisy channel
- a receiver
- a performance analyzer



Technical details

- all groups should use the same format for the txt files (format to be provided)
- a complete system is made of several transmitters, one channel and several receivers
- no carrier will be used (baseband)
- the transmitter, channel, receiver and performance analyzer must be implemented

Milestones

- Demo 1: week 04 (26.02.2025)
 - demonstration of a transmitter able to generate one single CDMA signal
 - input parameters: message length, spreading code length and number, spreading factor, sampling rate
 - demonstration of a channel able to combine multiple CDMA signals and add then noise
 - input parameters: signals file names, attenuation factors, noise level

The objective is to understand how CDMA signals are generated and how they are affected by the channel

Milestones

- Demo 2: week 08 (26.03.2025)
 - demonstration of a receiver able to decode multiple CDMA signals with noise
 - without synchronization concerns
 - plot of Bit Error Rate (BER) vs. Signal-to-Noise Ratio (SNR)

The objective is to evaluate the impact of the noise on the BER and the impact of having multiple users sharing the same channel.

Milestones

- Demo 3: week 13 (07.05.2025)
 - demonstration of a receiver able to decode one single CDMA signal with noise
 - with self-synchronization
 - demonstration of perfect detection for signals with high Signal-to-Noise Ratio (SNR)

The objective is to understand the need for synchronization.

Milestones

- Demo 4 + Final Presentation: week 17 (04.06.2025):
 - Demo of a receiver capable of decoding multiple CDMA signals with noise, including synchronization
 - Presentation of the final solution

The objective is to understand the impact of non-synchronized signals on the BER.

Milestones

- Final Report: week 18 (09.06.2025):
 - the final report should include:
 - a detailed description of the developed system
 - prints of the several waveforms
 - performance results for each one of the milestones
 - the used code (on a zip file)

Assessment summary

- Demo 1: 26.02.2025 (15%)
- Demo 2: 26.03.2025 (15%)
- Demo 3: 07.05.2025 (20%)
- Demo 4 + Final Presentation: 04.06.2025 (30%)
- Final Report: 09.06.2025 (20%)

Presence at demos and final presentation is mandatory.

File format

Line 1: sampled CDMA signal

Line 2: transmitted data 1 (sequence of bits)

Line 3: spreading code 1

Line 4: spreading factor 1

(...)

Line $3x(n-1)+2$: transmitted data n (sequence of bits)

Line $3x(n-1)+3$: spreading code n

Line $3x(n-1)+4$: spreading factor n

File format (example)

```
-1.03784,-1.06734,-1.04997,-0.049834,-0.0159142,0.0162229,0.0906242,-0.0731282,-0.0380791,0.073328,-  
0.0793481,0.0102576,0.0523822,1.01435,0.967696,1.01915,0.98632,0.972424,-0.0981943,-  
0.0855323,0.0219333,-0.0899182,-0.085237,0.00619708,-0.052677,0.935302,0.95423,1.04794,-0.904322,-  
0.930531,-1.09091,-1.04102,-1.08093,0.918765,0.928701,0.934409,1.00089,0.988435,-0.977781,-1.08592,-  
0.948985,-1.07059,-1.09227,1.07497,0.921658,0.919508,0.981575,1.03405,-0.938526,-1.08935,-  
0.0623497,0.0653034,-0.0997687,0.0755288,0.0753248,  
(...)  
-0.0284205,0.0927299,0.0460462,0.0634256,0.0690751,0.0491581,0.014122,-0.925704,-0.997962,-1.01767,-  
0.989227,-1.00654,-0.0641727,-0.093283,-0.00419789,-0.017165,-0.0305928,0.0301381,-  
0.0277701,0.0370131,-0.0253794,0.067912,-0.0347969,-0.0899309,-0.00452969,0.0403189,-0.0574997,-  
0.0869024,0.00769641,-0.0712145,0.00132445,0.0778868,-0.959555,-1.0928,-1.06558,-1.06927,-  
1.02095,0.0833072,0.0757037,0.00720031,0.0580433,0.077418,0.928199,-1.02338  
1,0,1,1,1,1,0,1,0,0,1,1,0,1,1,1,1,1,0,0  
-1,1,-1,-1,1,-1,1,-1,1,-1,1,1  
5  
1,1,1,0,0,1,0,0,1,1,0,0,1,1,0,1,1,0,0,0  
1,1,1,-1,-1,1,1,1,-1,1,-1,1,-1  
5
```